

NAMSSKOGAN, Norway

WMO: 012810

Lat: 64.7419N

Lon: 12.8458E

Elev: 140

StdP: 99.65

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.0	(c) -19.5	(d) -25.8	(e) 0.4	(f) -22.8	(g) -22.3	(h) 0.5	(i) -19.3	(j) 9.9	(k) -4.6	(l) 8.7	(m) -4.2	(n) 1.4	(o) 40	(p) 0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.3	(c) 26.6	(d) 16.7	(e) 24.1	(f) 15.8	(g) 21.8	(h) 14.6	(i) 17.8	(j) 24.9	(k) 16.5	(l) 22.7	(m) 15.3	(n) 20.7	(o) 2.9	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.0	10.8	20.2	13.6	9.9	19.1	12.5	9.2	17.7	50.5	25.0	46.6	22.8	43.2	20.8	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.5	6.4	5.5	-26.1	29.4	4.3	2.4	-29.2	31.1	-31.7	32.5	-34.2	33.8	-37.3	35.6
			-26.3	19.4	4.3	1.6	-29.4	20.6	-31.9	21.5	-34.3	22.4	-37.4	23.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-5.5	-5.2	-2.4	2.3	7.3	11.4	15.0	13.2	9.2	3.6	-1.2	-4.0		
	DBStd	8.41	6.33	6.12	5.10	3.16	3.99	3.62	3.61	3.06	3.01	3.37	4.97	5.76		
	HDD10.0	2676	482	427	383	230	106	24	2	6	48	198	337	433		
	HDD18.3	5361	741	660	641	480	343	211	117	162	273	456	587	691		
	CDD10.0	374	0	0	0	1	22	65	155	106	26	1	0	0		
	CDD18.3	17	0	0	0	0	1	2	12	2	0	0	0	0		
	CDH23.3	287	0	0	0	0	13	54	184	36	0	0	0	0		
	CDH26.7	62	0	0	0	0	2	8	50	2	0	0	0	0		
(14) Wind	WSAvg	2.6	2.9	2.9	2.8	2.5	2.7	2.5	2.4	2.3	2.4	2.4	2.5	2.5		
(15) Precipitation	PrecAvg	1290	137	114	111	79	58	69	88	90	125	141	123	151		
	PrecMax	1749	278	305	257	184	99	164	208	154	316	224	265	277		
	PrecMin	847	28	5	8	26	19	0	33	36	38	57	16	45		
	PrecStd	208	69	72	70	41	21	39	40	33	62	47	71	76		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.3	5.8	9.2	16.9	24.8	27.9	31.1	26.7	21.2	14.0	9.1	6.6		
		MCWB	4.0	3.7	4.0	9.5	14.6	16.2	19.2	17.6	14.2	10.2	8.0	4.8		
	2%	DB	3.9	4.7	6.4	12.8	21.7	24.8	27.8	24.1	18.2	11.9	7.4	4.9		
		MCWB	2.6	3.1	3.3	6.9	12.9	14.7	17.8	16.3	12.8	8.8	5.9	3.3		
	5%	DB	2.6	3.4	4.8	10.1	18.5	21.6	25.2	21.8	16.1	10.0	5.6	3.6		
		MCWB	1.3	1.9	2.7	5.4	11.4	13.6	16.8	15.3	12.0	8.0	3.9	2.2		
	10%	DB	1.5	2.1	3.8	7.6	15.1	18.9	22.6	19.7	14.5	8.3	4.2	2.6		
		MCWB	0.4	0.7	2.0	4.1	9.6	12.5	15.5	14.4	11.1	6.6	2.8	1.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	4.2	5.0	9.6	15.2	17.3	20.7	18.4	14.7	10.5	8.4	5.2		
		MCDB	5.0	5.3	6.6	16.2	24.1	25.9	28.4	24.8	18.9	13.0	9.1	6.0		
	2%	WB	2.9	3.2	3.9	7.5	13.4	15.7	18.6	16.9	13.6	9.1	6.1	3.6		
		MCDB	3.8	4.4	5.8	12.3	20.4	22.7	25.7	22.8	17.5	11.4	7.4	4.5		
	5%	WB	1.5	2.1	3.1	5.8	11.6	14.3	17.3	15.8	12.4	8.0	4.0	2.3		
		MCDB	2.4	3.3	4.9	9.1	18.1	20.2	24.3	20.9	15.5	9.7	5.2	3.3		
	10%	WB	0.5	0.9	2.0	4.5	9.8	12.9	16.0	14.7	11.3	6.8	2.9	1.4		
		MCDB	1.3	2.0	3.6	7.1	14.5	18.4	21.7	19.1	14.0	8.2	4.1	2.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.7	9.0	8.9	10.1	9.9	10.3	10.1	8.3	6.0	5.0	6.2		
		MCDBR	5.0	5.8	8.2	13.9	16.0	15.9	16.2	14.6	11.7	8.4	5.3	6.0		
		MCWBR	4.3	4.6	5.8	8.7	8.9	7.9	7.7	7.3	6.2	4.4	5.0			
	5% WB	MCDWR	4.8	5.7	7.5	11.8	15.0	13.9	14.4	12.6	10.7	6.9	5.2	5.8		
		MCWBR	4.3	4.8	5.6	7.7	8.5	7.1	7.3	7.1	7.0	5.4	4.5	5.0		
(40) Clear-Sky Solar Irradiance	taub	0.202	0.274	0.305	0.334	0.340	0.340	0.340	0.348	0.336	0.317	0.295	0.348	N/A		
	taud	1.941	2.153	2.196	2.257	2.397	2.476	2.478	2.501	2.543	2.405	2.651	N/A			
	Ebn at Noon	322	618	768	835	868	877	857	835	777	613	309	N/A			
	Edh at Noon	24	58	87	104	100	95	92	82	64	46	22	N/A			
(44) All-Sky Solar Radiation	RadAvg	0.11	0.55	1.82	3.64	4.60	4.74	4.52	3.47	1.90	0.78	0.19	0.04			
	RadStd	0.01	0.10	0.30	0.37	0.33	0.58	0.56	0.39	0.23	0.10	0.03	0.00			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-142	-178	N/A	N/A

Nomenclature: See separate page